

VDAC2 STABILIZES A MEMBRANE-INSERTED, PRIMED INTERMEDIATE OF BAX ACTIVATION

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ABSTRACT

The Bcl-2 associated X protein (BAX) is a major effector of mitochondrial apoptosis implicated in a wide range of cancers. It is activated through a series of conformational transitions that lead to mitochondrial outer membrane permeabilization. Genetic studies have established the mitochondrial porin VDAC2 as an essential regulator of BAX-mediated apoptosis, yet the molecular basis of this regulation remains unresolved. The absence of direct structural and biochemical characterization of VDAC2–BAX interactions has prevented mechanistic understanding of how VDAC2 influences BAX activation. Here, using complementary biochemical, biophysical and structural approaches, we reconstituted and characterized, for the first time, a stable VDAC2–BAX complex that was not observed with VDAC1, highlighting an isoform-specific role in BAX regulation. We show that VDAC2 captures and stabilizes a primed BAX conformation displaying the hallmarks of BAX activation on mitochondria, including membrane insertion, BH3-domain exposure, and increased disorder of the N-terminal region. By integrating AlphaFold3 predictions with molecular dynamics simulations, biochemical, biophysical and structural constraints, we derive an experimentally-supported structural model in which BAX is anchored with VDAC2 in lipid nanodiscs through its $\alpha 9$ helix, while its core domain extends flexibly over the VDAC2 pore. Together, our findings support a model in which VDAC2 facilitates BAX membrane insertion and stabilizes a

membrane-inserted, activation-competent BAX intermediate. Rather than serving as a structural component of apoptotic pores, VDAC2 acts as a regulatory checkpoint in BAX activation, trapping a ‘primed’ intermediate of the BAX activation pathway. These results provide a molecular explanation for the emerging role of VDAC2 in mitochondrial apoptosis and establish structural basis for a previously inaccessible intermediate in the BAX activation pathway.

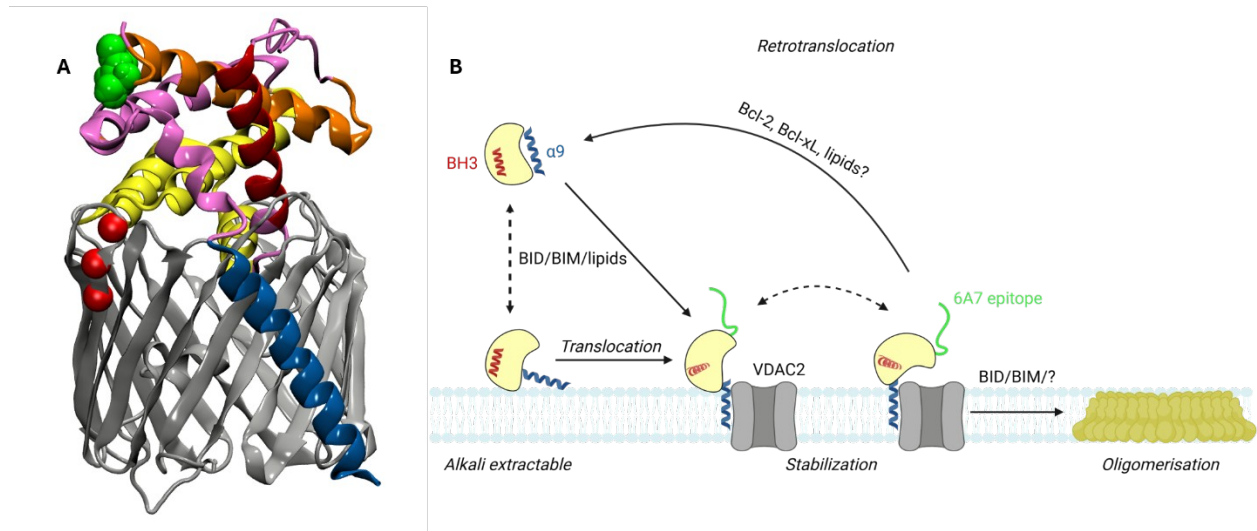


Fig. Structural model and functional role of the VDAC2-BAX complex

A. Integrative structural model of the VDAC2-BAX complex using MD simulations, cysteine crosslinking, XL-MS, cryo-EM, electrophysiology, mutagenesis, and antibody accessibility measurements. VDAC2 is shown in grey, and BAX is coloured with its different domains.

B. Proposed role of VDAC2 during BAX activation. Following membrane engagement and α9 insertion, VDAC2 captures and stabilizes a membrane-inserted, activation-competent BAX state. This intermediate may subsequently progress toward BAX oligomerization and apoptotic pore formation or undergo deactivation and retrotranslocation.